

Sample Project Documentation

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1. Introduction

Smart Lender is an AI/ML-based web application that predicts loan eligibility for applicants using historical lending data. The project was developed as part of the SkillWallet Group Project module under the Artificial Intelligence and Machine Learning track.

2. Objective

To automate the initial loan eligibility assessment process, reducing decision time and improving consistency compared to fully manual review.

3. Methodology

- Data Collection: Historical loan application dataset.
- EDA: Univariate, Bivariate, and Multivariate analysis to uncover patterns.
- Pre-processing: Missing value handling, SMOTE balancing, feature scaling, train/test split.
- Modeling: Decision Tree, Random Forest, KNN, and XGBoost compared on standard metrics.
- Deployment: Best model wrapped in a Flask web application with a simple HTML interface.

4. System Architecture

User Browser to HTML Form to Flask Backend to Trained ML Model to Prediction to Result Page.

5. Results Summary

Random Forest achieved the best overall performance among the models tested and was selected for deployment in the Flask web application.

6. Conclusion

Smart Lender demonstrates that a lightweight ML pipeline combined with a simple web interface can meaningfully speed up and standardize the loan eligibility pre-check process.

7. Future Scope

- Add loan amount recommendation, not just approve/reject classification.
- Add explainability (e.g. SHAP values) so users understand which factors influenced their result.
- Integrate with real banking data sources under proper data-governance controls.